

Bayesian estimation of von Bertalanffy growth parameters for gray triggerfish, *Balistes capriscus*, incorporating multiple readers and ageing structures

Derek W. Chamberlin^{1a,b}, Zachary A. Siders^a, Jennifer C. Potts^c, Walter D. Rogers^d, Miaya A. Taylor^a, and William F. Patterson, III^a

^aUniversity of Florida, Fisheries and Aquatic Sciences, 7922 NW 71st Street, Gainesville, FL 32611, USA; ^bNational Marine Fisheries Service, Alaska Fisheries Science Center, 7600 Sand Point Way N.E., Building 4, Seattle, WA 98115, USA; ^cNational Marine Fisheries Service, Southeast Fisheries Science Center, Beaufort Laboratory, 101 Pivers Island Rd, Beaufort, NC 28516, USA; ^dCooperative Institute for Marine and Atmospheric Studies, University of Miami, in support of NOAA Fisheries Southeast Fisheries Science Center, Beaufort Laboratory, 101 Pivers Island Rd, Beaufort, NC 28516, USA

Corresponding author: Derek W. Chamberlin (email: derek.chamberlin@noaa.gov)

Abstract

Ageing error can significantly impact growth parameter estimates. However, ageing error is rarely considered as a source of uncertainty as growth is generally estimated from a single age reader's age estimates. Here, we developed a sex-specific Bayesian hierarchical formulation of the von Bertalanffy growth model (VBGM) to estimate gray triggerfish (*Balistes capriscus*) growth parameters from multiple ($n = 3$) reader age estimates derived from otolith opaque zone counts, a new dorsal spine protocol, and the historical (old) dorsal spine protocol. For all three ageing protocols, estimates of length-at-age were significantly different between sexes, resulting in significantly higher estimates of L_{∞} for males compared to females. Estimates of the Brody growth coefficient (k) from both the otolith and new spine ageing protocols were significantly lower than the old spine protocol estimate. Study results provide a framework for comparing VBGMs, suggest it is most appropriate to fit sex-specific growth models for this species, and suggest stock assessments using ages estimated with the old spine ageing protocol would produce biased results.

Key words: gray triggerfish, von Bertalanffy, Bayesian, age and growth

Introduction

Accurate estimation of growth parameters is fundamental to the accuracy of many fisheries stock assessment models. Size-structured stock assessment models rely on estimates of growth to determine the progression of fish across timesteps, while age-structured models frequently use growth to calculate total biomass or biomass-at-age and often outperform their size-structured counterparts (Fisch et al. 2019). Proxy estimates of fecundity, mortality, and vulnerability are also frequently derived from growth estimates, which are also critical to determining how size limits or changes in selectivity may affect yield (von Bertalanffy 1934; Walters and Martell 2004; Zhou et al. 2012; Stawitz et al. 2019). Thus, accurate estimation of growth parameters, and the attendant uncertainty in those estimates, is fundamental to accurate stock assessment outputs and effective fisheries management.

Growth estimation has advanced from fitting growth functions solely to age and length data (Quinn II and Deriso 1999), to incorporating multiple readers as random effects (Cope and Punt 2007) and considering marginal increment data or multiple readers within a Bayesian framework (Caltabellotta

et al. 2021; Chamberlin et al. 2023b). Early implementation of growth models estimated growth parameters without uncertainty parameters (von Bertalanffy 1934). Under that framework, growth estimates represent mean growth, across the samples to which the model is fit, and process error (natural variability) and observation error (ageing error) were not estimated. As statistical and computational capabilities have improved uncertainty in parameter estimates is typically estimated. When uncertainty is estimated, nonlinear least squares or likelihood formulations typically assume that variance between observations and the model is due solely to variability in length-at-age, or process error (Quinn II and Deriso 1999; Cope and Punt 2007), and fail to consider how observation error may contribute to the overall variance structure. By incorporating multiple age estimates the relative contribution of observation error to the total variance can be quantified (Cope and Punt 2007; Chamberlin et al. 2023b) and the total variance can subsequently be incorporated into stock assessments.

Gray triggerfish (*Balistes capriscus*) in the Gulf of Mexico (GOM) is a stock for which there has historically been

substantial ageing error (Potts et al. 2023; Chamberlin et al. 2024). This may have contributed to the failure of the stock to recover at expected rates following decades of overfishing and the implementation of a rebuilding plan in the early 1990s (SEDAR 2006, 2015). Gray triggerfish were historically aged by counting translucent zones in dorsal spine sections using the methods of Kolmos et al. (2013), hereafter referred to as the old spine protocol. However, this method results in biased underestimates of age (Potts et al. 2023; Chamberlin et al. 2024). Underestimation of age results in biased estimates of growth, mortality, recruitment, and the overestimation of the fishing mortality that produces the maximum sustainable yield (Beamish and McFarlane 1983; Lin Lai and Gunderson 1987; Tyler et al. 1989; Bradford 1991; Campana 2001; Reeves 2003). Therefore, considering ageing error (i.e., bias and imprecision) as a source of uncertainty is essential to effectively characterizing parameters derived from age data, particularly in the case of gray triggerfish.

Potts et al. (2023) recently reported the annual periodicity of growth zone formation in gray triggerfish dorsal spines and otoliths and proposed a modified protocol for estimating age from dorsal spine thin sections, referred to as the new spine protocol. Chamberlin et al. (2024) later validated age estimates derived from whole otoliths and the Potts et al. (2023) new spine protocol as being accurate via application of the bomb radiocarbon chronometer to eye lens core (birth year) $\Delta^{14}\text{C}$ values. However, it is also important to consider how differences in between-reader precision affect parameter estimates derived from either ageing structure as between-reader precision differs between the two protocols. Additionally, it is important to consider how growth parameters estimated from these validated methods compare to estimates derived from the biased method of the Kolmos et al. (2013), which historically was used to estimate age as well as growth parameters for use in GOM gray triggerfish stock assessments (e.g., Allman et al. 2016, 2018). However, the direct comparison of growth parameter estimates is complicated by the effects of samples sizes and presence or absence of small or large fish (Gwinn et al. 2010; Wilson et al. 2015). Bayesian methods allow parameters from different models to be estimated simultaneously and can consider the joint distribution across correlated growth parameters, allowing for a direct comparison of parameter estimates between different models, which is critical in estimating and comparing growth under alternate ageing protocols.

The objectives of this study were to develop a modeling framework to (1) estimate GOM gray triggerfish growth parameters from age estimates produced by multiple readers; (2) test for differences in growth parameters computed from age estimates derived from whole otoliths versus those produced from the new spine-based method reported by Potts et al. (2023) and the old method of Kolmos et al. (2013); and (3) test whether gray triggerfish exhibit sexually dimorphic growth. Study results are discussed in the context of their implications for estimating gray triggerfish growth and population dynamics, as well as broader applicability of the methods presented here to fish growth estimation.

Methods

Sample collection

Fishery-independent and fishery-dependent gray triggerfish samples were collected in the GOM with hook and line, longline, trawl, and spear. Animal collection was permitted by the National Marine Fisheries Service and animals were handled in accordance with the Guide for the Care and Use of Laboratory Animals under protocols approved by the University of Florida Institutional Animal Care and Use Committee (Protocol #202211613). Fishery-dependent and independent samples collected with hook and line were targeted on natural and artificial reefs in the northern Gulf of Mexico (nGOM) off Dauphin Island, Alabama to Destin, Florida, and the eastern Gulf of Mexico (eGOM) on the west Florida continental shelf (WFS). Fishery-dependent samples collected with longline were targeted by commercial fishers on natural and artificial reefs also on the WFS. Trawl caught samples were collected throughout the GOM during the Southeast Area Monitoring and Assessment Program trawl survey. Finally, gray triggerfish that were speared were collected in the western Gulf of Mexico (wGOM) off the coast of Texas. Combining samples from multiple gear types serves to reduce bias and improve precision of growth estimates by increasing the sample's representativeness of the population at large (Wilson et al. 2015).

Fishery-dependent samples had been filleted and all gray triggerfish carcasses, from sampled offloads, were transported to the University of Florida on ice. Each carcass was measured to the nearest millimeter filleted fork length (FL). Sex was determined by macroscopic examination of the gonads; if the gonads had been removed sex was indicated as unknown. All fishery-independent gray triggerfish samples were transported to the University of Florida whole on ice to be processed. Each fish was measured to the nearest mm FL and weighed to the nearest gram (g). A subset of fishery-independent samples were subsequently filleted and the filleted FL was measured. A linear-regression was then fitted to the whole FL and filleted FL, which was used to convert filleted FL to whole FL for fishery-dependent samples. While length measurement error is a source of uncertainty, albeit small, (Bunch et al. 2013) we lack the repeated measurement data to fully estimate length measurement error; thus, we did not include this source of uncertainty in our models. Right and left vestibular systems were extracted whole, with otoliths contained within the saccule, and stored dry in 0.5 mL microcentrifuge tubes. The first dorsal spine, including the condyle, was removed by cutting anterior and posterior to the spine.

Age estimation and size-at-age

Sagittae were removed from vestibular tissue with tweezers in the laboratory following rehydration of the tissue, and then stored dry in microcentrifuge vials. Each otolith sample was read whole by an experienced reader (R1; DWC) and a subset of otoliths were read by two additional experienced fish agers (R2; JCP and R3; WDR) using the methods of Chamberlin et al. (2024). Dorsal spines were prepared

for ageing by boiling in water and subsequently scraping to remove any adhering tissue (Allman et al. 2016). A 0.5 mm thick transverse section was cut near the spine's base with a diamond-bladed, low-speed saw. The section was then affixed to a slide and covered with a toluene-based mounting medium. Translucent zones were initially enumerated by R1 using the old spine protocol (Kolmos et al. 2013). Second and third reads were not conducted using this protocol as R2 and R3 had developed a new spine protocol and reverting to the old spine protocol would likely have produced biased results. Following training by R2 and R3, translucent zones were also enumerated by R1 using the new spine protocol (Potts et al. 2023). A subset of these samples were also read by R2 and R3. All reads were performed double-blind, without knowledge of other reader's opaque or translucent zone counts or morphometric data.

The index of average percent error (iAPE) was calculated to assess the precision of growth zone counts between readers for each ageing structure (Beamish and Fournier 1981). Bias plots were also constructed to compare growth zone counts between readers for each ageing structure (Campana et al. 1995), as well as between ageing structures or spine protocols for each reader. Fractional age was estimated for age-0 fish (Frac₀) by dividing the measured FL by the estimated juvenile growth rate of 0.95 mm/day and then dividing by 365 days (eq. 1; Ingram 2001; Moore 2001).

$$(1) \quad \text{Frac}_0 = \frac{(\text{FL}/0.95)}{365}$$

The effects of sex on estimated size-at-age derived from R1 data were tested with three-factor ANOVA models ($\alpha = 0.05$) in R (R Core Team 2023). FL was the dependent factor, with age, ageing structure, and sex as independent factors. The ageing structure*age*sex interaction was sliced by ageing structure and age to test for sex-specific differences in size-at-age. All analyses were performed in R (R Core Team 2023).

Von Bertalanffy growth estimation

Von Bertalanffy growth model formulation

We used the size-at-birth formulation of the von Bertalanffy growth model (VBGM; von Bertalanffy 1934) as a template to build a growth model for gray triggerfish (eq. 2).

$$(2) \quad L_i = L_\infty - (L_\infty - L_0) e^{-kt_i}$$

where L_i is the length at some age t_i of individual i , L_∞ is the asymptotic length or average maximum size, L_0 is the size at birth, and k is the Brody growth coefficient. We adapted this growth model to (1) compare growth parameter estimates between ages derived from otoliths and both spine protocols; (2) compare growth parameters between sexes; (3) incorporate unknown sex individuals; and (4) account for between-reader variability. To compare among otolith-derived and old and new spine-derived ages, we fit separate instances of eq. 2 while sharing a common L_0 among models (eq. 3)

and specifying $L_{i,p}$ to be described by a log-normal process (eq. 4).

$$(3) \quad \hat{L}_{i,p} = L_{\infty,p} - (L_{\infty,p} - L_0) e^{-k_p t_{i,p}}$$

$$(4) \quad L_{i,p} \sim \text{logNormal}(\log(\hat{L}_{i,p}), \sigma_p)$$

where p denotes different protocols, i, p describes individual i aged with hard part/protocol p , $\hat{L}_i$ describes the predicted length, L_i now describes the observed length, and σ_p describes the standard deviation on log scale, the variability of measurements on the logarithmic scale. We then further generalized eqs. 3 and 4 to account for sex (eqs. 5 and 6).

$$(5) \quad \hat{L}_{i,p} = L_{\infty,p,s} - (L_{\infty,p,s} - L_0) e^{-k_{p,s} t_{i,p}}$$

$$(6) \quad L_{i,p} \sim \text{logNormal}(\log(\hat{L}_{i,p}), \sigma_{p,s})$$

where s denotes the sex-specific parameter. As we did not know the sex of every individual, we modified the sex-specific parameters such that the von Bertalanffy growth parameters are linear models with a shared intercept across sexes and then additive effects of sex (eqs. 7–9) in the spirit of Siders et al. (2023).

$$(7) \quad \log(L_{\infty,p,s}) = \log(L_{\infty,p}) + \beta_{L_{\infty},s}$$

$$(8) \quad \log(k_{p,s}) = \log(k_p) + \beta_{k,s}$$

$$(9) \quad \log(\sigma_{p,s}) = \log(\sigma_p) + \beta_{\sigma,s}$$

where $\beta_{X,s}$ is the sex-specific additive effect to a given VBGM parameter.

Lastly, we accounted for between-reader variability in the assigned age, which is a major source of error in variables in the age-growth modeling process. We adopted the hierarchical modeling approach of Chamberlin et al. (2023b) that uses the multiple reads as observations of the true age for each individual coming from a location-scale T distribution, a generalization of the Student's T distribution (eq. 10),

$$(10) \quad t_{i,p,r} \sim \text{location-scaleT}(\nu_{i,p}, \hat{t}_{i,p}, \tau_p)$$

where $t_{i,p,r}$ is the observed age coming from reader r per individual i and protocol p , $\nu_{i,p}$ is the degrees of freedom equal to the number of reads, $\hat{t}_{i,p}$ is the "true" age, and τ_p is the inter-reader variability. Including τ_p separates the ageing observation error from σ_p and $\sigma_{p,s}$ such that these now reflect process error or growth variability. A location-scale T distribution was used due to the incorporation of fractional ages for fish with 0 observed growth zones (age-0) and has the added benefit of accounting for the number of individual reads per sample.

Within the Bayesian VBGM, we predicted length-at-age for each sex and protocol over a range of hypothetical ages from zero to $t_{\max}$ by 0.1 year increments. For integration with the more common t_0 formulation of the VBGM in teleosts, we derived $\hat{t}_0$ (eq. 11),

$$(11) \quad \hat{t}_{0,p,s} = \frac{1}{k_{p,s}} \log\left(\frac{L_{\infty,p,s} - L_0}{L_{\infty,p,s}}\right)$$

Von Bertalanffy growth model implementation

We used STAN (Stan Development Team 2022) and *cmdstanr* (Gabry and Češnovar 2022) to implement the Bayesian VBGM for gray triggerfish. To do so, we first fit eq. 2 using maximum-likelihood estimation to estimate L_∞ , k , and σ and then, following Caltabellotta et al. (2019), Chamberlin et al. (2023b), and Rolim et al. (2020), generated weakly informative priors using the predicted mean for the prior location and a 50% coefficient of variation to generate the corresponding scale. For L_∞ , k , σ , and τ , we specified normal priors on the log-transformed parameters (Table S1). A weakly-informative normal prior was put on $\log(\tau)$ with a mean of -1.0 and a standard deviation of 4.0 (Table S1). The only parameter that received a strongly informative prior was L_0 with a mean of 2.2 mm and standard deviation of 0.04 mm, reported by Simmons and Szedlmayer (2013). For the sex-specific additive effects, we specified normal priors with mean 0 and a standard deviation of 1 (Table S1) as these effects were on the log-transformed scale (eqs. 7–9). We used the No U-Turn sampler in STAN with eight chains that had 5000 warmup iterations and 125 sampling iterations each. We combined the chain-specific posterior samples to generate a single posterior of each parameter after assessing chain coverage using the Gelman and Rubin (1992) statistic below 1.1 for all parameters. We assessed significant differences between parameters using the maximum *a posteriori* (MAP) based p -value (p_{MAP}) at $\alpha = 0.10$ using the bayestestR package (Makowski et al. 2019a, 2019b). Additionally, we conducted a sensitivity run with hyper weakly informed priors, from Jefferson et al. (2019), placed on L_∞ , and k while retaining the informative prior on L_0 (Table S2).

Results

Sample collection

In total, 1270 gray triggerfish were samples across the GOM, consisting of 925 fishery-dependent samples and 345 fishery-independent samples (Figs. S1 and S2). Of these, 706 samples were collected in the nGOM, 391 in the eGOM, and 122 in the wGOM (Fig. S1). Among fishery-dependent samples, 51 did not have a sample location noted. There were 685 fish sampled with hook and line, 412 with longline, 122 with spear, and 51 with trawl. There was a significant relationship between whole FL and filleted FL ($n = 42$; $R^2 = 0.99$; $\text{FL}_{\text{Whole}} = 0.99 * \text{FL}_{\text{Filleted}} + 0.30$), which was used to convert filleted FL to whole FL for fishery-dependent samples (Fig. S3).

Age estimation and size-at-age

Age was estimated by R1 for 1182 samples by counting whole otolith opaque zones, 1219 samples using the new spine protocol (Potts et al. 2023), and 1243 samples using the old spine protocol (Kolmos et al. 2013). Differences in sample sizes are due to missing structures or differences in interpretability. Between-reader growth zone counts derived from whole otoliths and the new spine protocol showed strong agreement among readers (see Chamberlin et al. 2024; Fig. 3 for reader bias plots). Overall, iAPE among all three read-

ers was 10.1% for whole otolith-derived age estimates and 9.4% for new spine protocol age estimates. Therefore, age estimates were slightly more consistent between readers using the new spine protocol compared to reading whole otoliths (Chamberlin et al. 2024). There are no precision metrics for the old spine protocol as only R1 read the dorsal spine samples using that protocol.

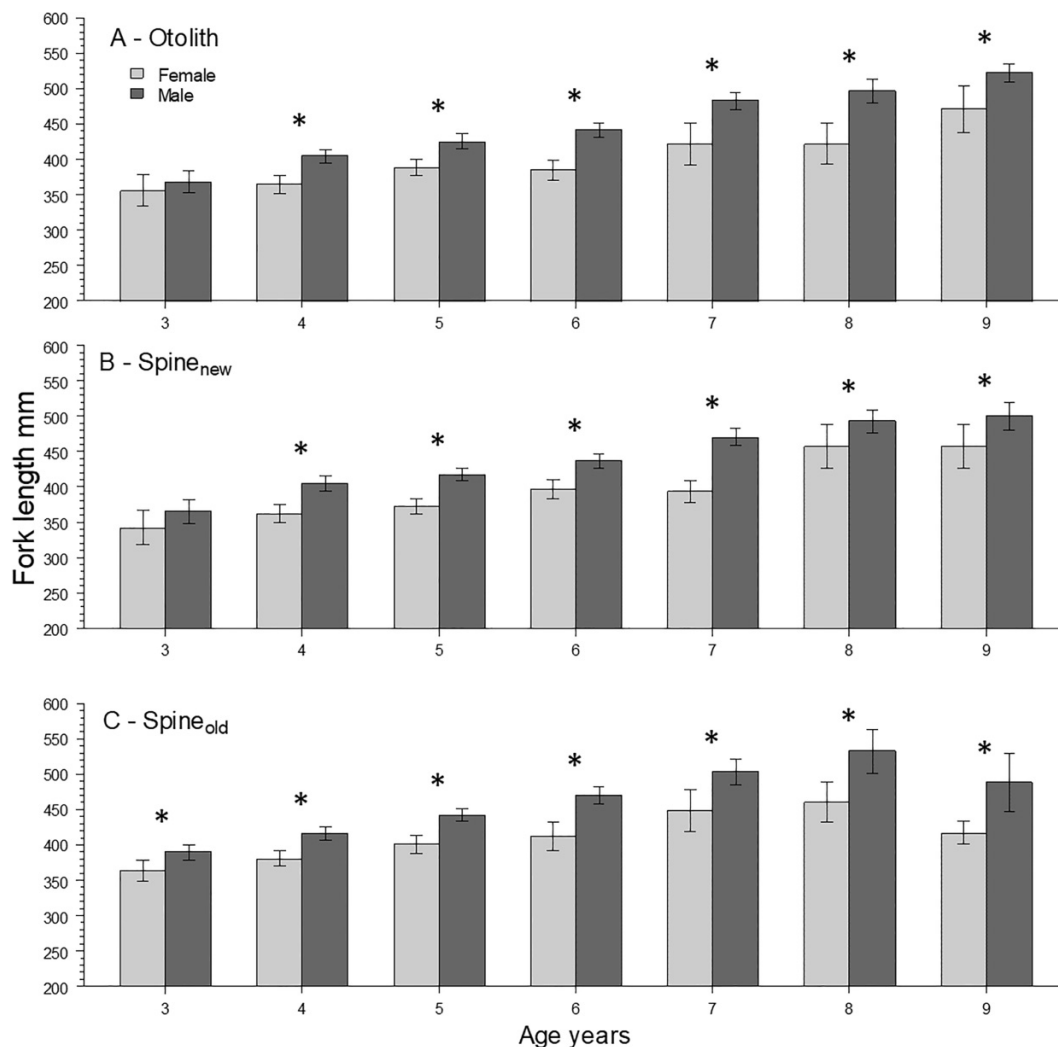
Otolith-derived age estimates ranged from 0.40 to 14 years for female gray triggerfish and 0.43 to 12 years for males. New spine protocol age estimates ranged from 1 to 16 years for female gray triggerfish and 0.37 to 14 years for males, while old spine protocol age estimates ranged from 1 to 12 years for female gray triggerfish and 0.37 to 9 years for males. Otolith and new spine protocol age estimates corresponded well across ages with no apparent relationship between sex and protocol-specific age estimates (Fig. S4). Conversely, age estimates derived from the old spine protocol were consistently less than those derived from the validated whole otolith and the new spine protocols, for ages ≥ 5 (Fig. S4). Males were significantly larger at age than females when age was estimated from either whole otoliths, the new spine protocol, or the old spine protocol (Fig. 1; Table S3).

Von Bertalanffy growth estimation

The Bayesian hierarchical VBGM with size-at-age data from all three ageing protocols converged (Figs. S5–S9) with no Gelman and Rubin (1992) statistic over 1.01 , mean bulk effective sample sizes of 1811, and mean tail effective sample sizes of 2488 (Table S4). Among all models two parameters, $\sigma[\text{spine old, M}]$ and $\sigma[\text{spine old, U}]$, had a bulk effective sample size of 212 and 220, respectively, which suggests that while the central tendency of the posterior is robust the tail probabilities for these two parameter estimates may be poorly sampled. Differences between parameter estimates from the model run with weakly informative priors and the sensitivity run were restricted to differences in the last significant digit (Table S5).

Differences between the von Bertalanffy parameter estimates based on otolith-, new spine protocol-derived ages, or old spine protocol-derived ages were insignificant, with the exception of k and t_0 in the sex-specific models (Figs. 2 and 3; Table S6). At the median k for the combined sex model, the differences between otolith- ($k = 0.42 \text{ year}^{-1}$) and new spine protocol-derived ($k = 0.39 \text{ year}^{-1}$) parameter estimates were only 0.03 year^{-1} (Table 1). However, the combined sex old spine protocol estimate ($k = 0.55 \text{ year}^{-1}$) was greater than estimated k in either the combined sex otolith or new spine protocol models. This difference was not significant due to variability in posterior estimates, which likely resulted from sexually dimorphic growth (Table S6). The greater between-reader variability for otolith-derived ages compared to new spine protocol-derived ages was not statistically significantly (Table S6, $p_{\text{MAP}} = 0.52$ of $\Delta\tau$). Inclusion of multiple readers resulted in increased uncertainty in the combined sex otolith and new spine protocol models ($\sigma_{\text{total}} = \sigma + \tau = 0.68$ for both the otolith and new spine protocols, respectively) compared to the single reader old spine model ($\sigma_{\text{total}} = \sigma = 0.14$).

Fig. 1. (A) Mean sex-specific size-at-age ($\pm 95\%$ CI) for gray triggerfish for otolith-derived age estimates for ages 3–9 years. Age-specific sample sizes were 86, 161, 185, 170, 76, 52, and 41 for ages 3–9 years, respectively. (B) Mean sex-specific size-at-age ($\pm 95\%$ CI) for gray triggerfish for new spine protocol-derived age estimates for ages 3–9 years. Age-specific sample sizes were 76, 144, 180, 195, 92, 57, and 43 for ages 3–9 years, respectively. (C) Mean sex-specific size-at-age ($\pm 95\%$ CI) for gray triggerfish for old spine protocol-derived age estimates for ages 3–9 years. Age-specific sample sizes were 115, 226, 247, 136, 62, 18, and 6 for ages 3–9 years, respectively. Asterisks denote a significant ($\alpha < 0.05$) effect of sex within year class.



Differences between protocols were more pronounced in the sex-specific models where female- and male-specific old spine protocol-derived estimates for k and t_0 significantly differed from the otolith and new spine protocols (Fig. 3; Table S6). For k , these estimates were, across otolith and new spine protocol-derived estimates, 0.18 year^{-1} higher for females and 0.13 year^{-1} higher for males, on average, compared to the old spine protocol-derived estimates (Table 1). The significant differences in k were driven by the relative underestimation of age using the old spine protocol compared to the otolith or new spine protocols (Fig. S4). While there were significant differences in estimates of $\hat{t}_0$, this was due to the strong prior on L_0 resulting in narrow posterior distributions of $\hat{t}_0$ (Table 1).

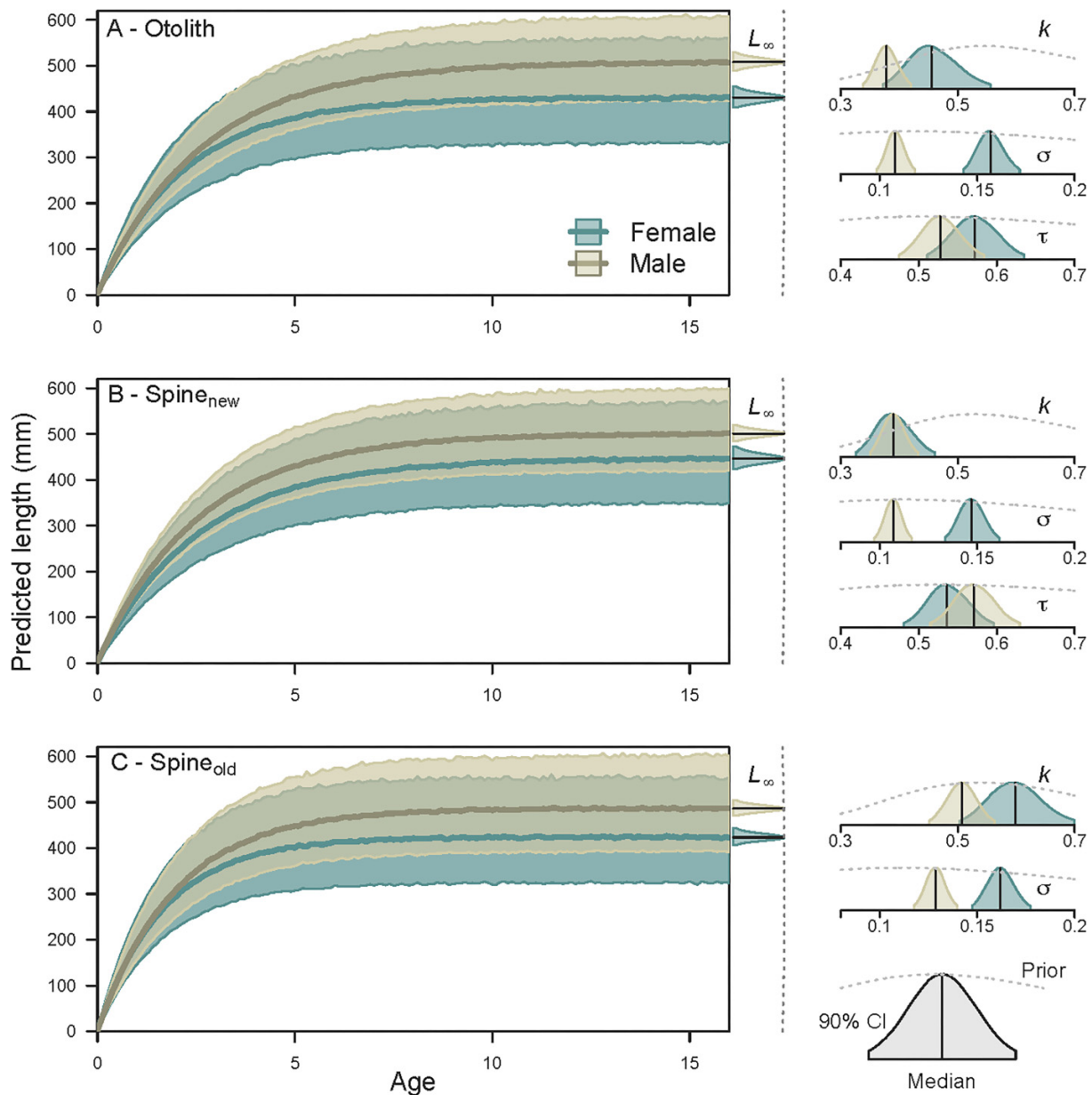
In terms of sex-specific growth, males were expected to grow significantly larger than females ($L_{\infty, M} = 502 \text{ mm}$, $L_{\infty, F} = 447 \text{ mm}$, new spine protocol-derived; $p_{\text{MAP}} < 0.01$) but both

sexes are roughly the same size until age 2 to age 3 (Figs. 1, 2, and 4; Table 1). Females had significantly higher variability in growth (Fig. 4; $p_{\text{MAP}} < 0.01$ of $\Delta\sigma$). Females generally had higher estimates of k but not significantly so due to uncertainty in the posterior parameter estimates of k regardless of otolith- or spine-derived ages (Fig. 4; Table 1; $p_{\text{MAP}} = 0.18\text{--}0.98$). Otolith-derived estimates of k were more different between sexes than the new spine protocol-derived ages (Fig. 4), due to greater between-reader variability and a tendency for female otolith-derived ages to be slightly lower at younger ages than the corresponding female new spine protocol-derived ages (Fig. S4).

Discussion

Our results demonstrate that gray triggerfish age estimates derived from whole otoliths and the Potts et al. (2023)

Fig. 2. Sex-specific von Bertalanffy growth models fitted to (A) otolith-derived age estimates, (B) dorsal spine-derived age estimates using the new Potts et al. (2023) protocol, and (C) dorsal spine-derived age estimates using the old Kolmos et al. (2013) protocol. Data from multiple readers were utilized in otolith and new spine growth models. The median (solid line) and 90% credible intervals (CIs) (shaded areas) are shown for sex-specific growth model predictions in each panel. Posterior distributions, median estimate (solid, vertical line), 90% CIs, and priors (dashed line) are shown for asymptotic length (L_{∞}), the Brody growth coefficient (k), and the likelihood standard deviations (σ and τ).

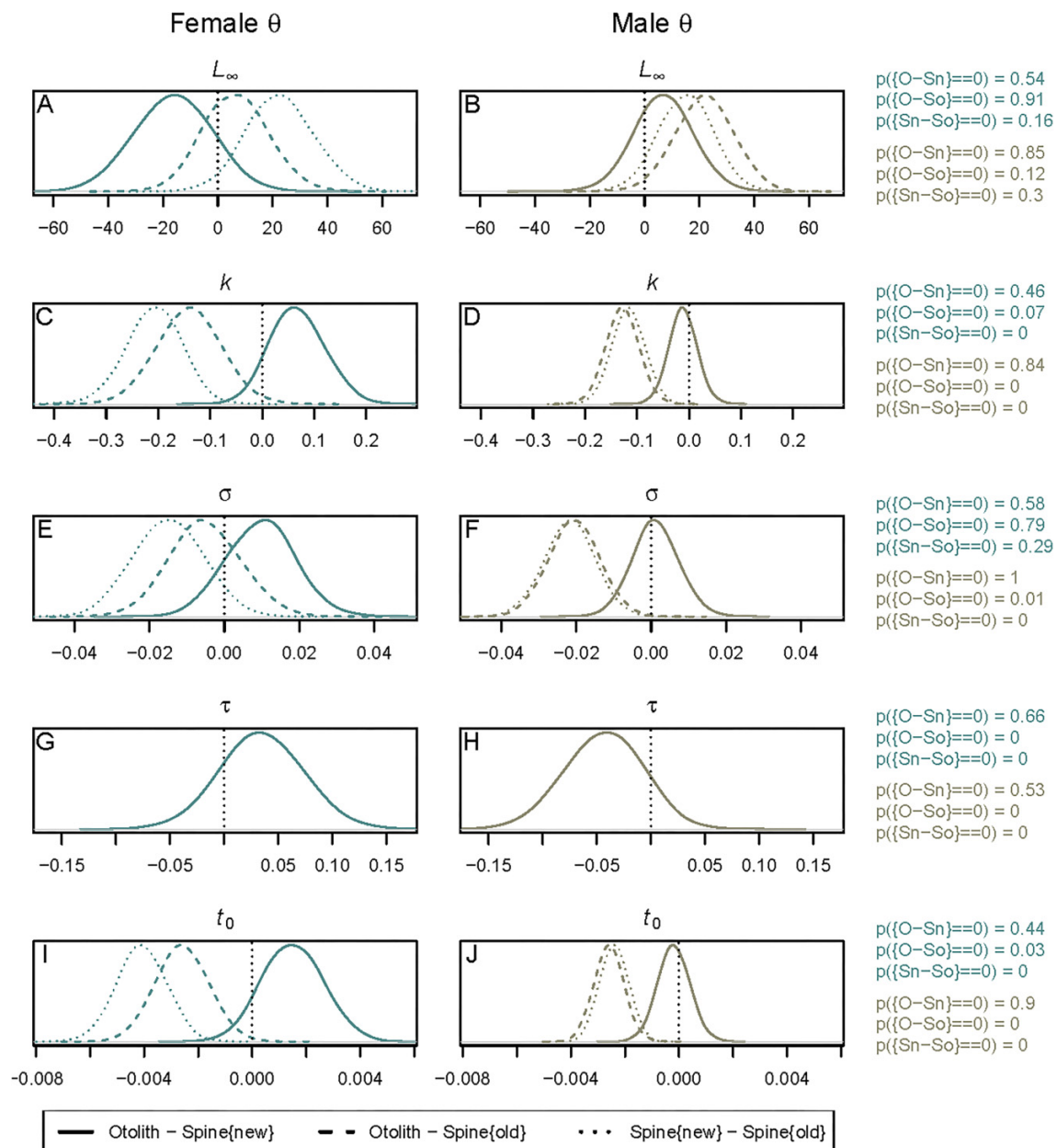


dorsal spine protocol produce similar growth parameter estimates, while age estimates using the Kolmos et al. (2013) dorsal spine protocol result in a significantly higher estimate of the Brody growth coefficient, k . The Brody growth coefficient is a common proxy for natural mortality (M) that is a proxy for stock productivity; thus, a higher k could be indicative of high M indicating a more productive stock; thus, overestimating k could result in overestimating stock productivity (Pauly 1980; Walters and Martell 2004). Additionally, our results clearly demonstrate gray triggerfish exhibit sexually dimorphic growth with males growing to a larger

maximum size than females on average. This suggests future assessments should consider incorporating sex-specific estimates of growth and landings data to better capture the true population dynamics.

Growth misspecification can bias stock assessment estimates resulting in overly optimistic estimates of management reference points (Stawitz et al. 2019), which increases the likelihood of overfishing. Increasingly stock assessments are able to incorporate uncertainty parameters and propagate that uncertainty into assessment outputs (Methot et al. 2020). Incorporating multiple readers into growth estimation

Fig. 3. Posterior distribution of the difference, by sex, in asymptotic length (L_{∞} ; A–B), the Brody growth coefficient (k ; C–D), the likelihood standard deviations (σ ; E–F), the likelihood standard deviation due to observation error (τ ; G–H), and size at length zero (t_0 ; I–J) estimates between age estimation methods for otolith- and new spine protocol- (solid line), otolith- and the old spine protocol- (dashed line), and new spine protocol- and old spine protocol- (dotted line) derived age estimates. The maximum *a posteriori* (MAP) value is reported for ageing method comparison to the right of each panel with significant differences assessed at $\alpha = 0.10$.



and estimating observation error allows for sources of process error, such as annual or spatial growth variability, to be investigated explicitly (Cope and Punt 2007). While process error would generally be reduced, the estimation of observation error results in an overall increase in total uncertainty because the parameter estimates now incorporate ageing error. The increase in overall uncertainty would have attendant effects by increasing uncertainty in any derived parameter, such as M or fecundity, whereas under a single reader model this uncertainty could not be captured and accounted for.

Given these advantages future growth estimation should explicitly acknowledge and incorporate ageing error by estimating growth parameters based on multiple reads.

Direct comparison of growth parameter estimates is difficult as parameter estimates are affected by samples size and the presence or absence of small and/or large fish (Gwinn et al. 2010; Wilson et al. 2015). Alternatively, differences in size-at-age can be tested directly (Shepard et al. 2010; Chamberlin et al. 2023a). However, the direct comparison of growth parameter estimates is often necessary to determine

Table 1. Posterior median estimates and 90% credible intervals for gray triggerfish von Bertalanffy growth model parameters.

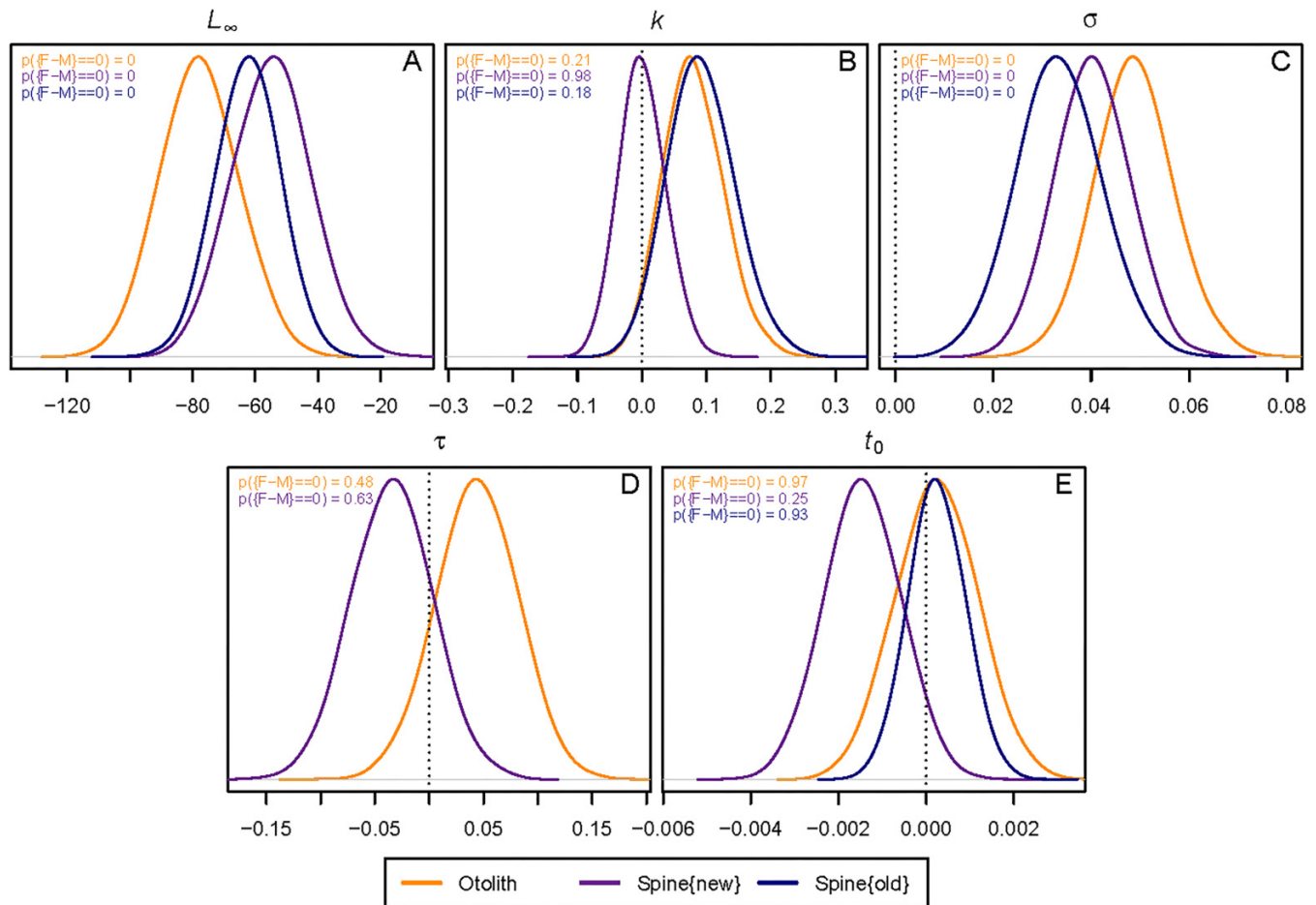
Protocol	Sex	θ	Median	90% credible interval
Otolith	Combined	L_{∞}	468	458–478
		k	0.416	0.384–0.454
		$\hat{t}_0$	–0.012	–0.013 to –0.011
		σ	0.130	0.124–0.136
		τ	0.55	0.52–0.58
Spine—new	Combined	L_{∞}	473	464–484
		k	0.389	0.363–0.42
		$\hat{t}_0$	–0.012	–0.013 to –0.012
		σ	0.125	0.120–0.132
		τ	0.554	0.525–0.584
Spine—old	Combined	L_{∞}	454	446–463
		k	0.550	0.513–0.592
		$\hat{t}_0$	–0.009	–0.010 to –0.009
		σ	0.144	0.138–0.151
		τ	—	—
Otolith	Female	L_{∞}	431	416–446
	Male	L_{∞}	508	496–522
	Female	k	0.456	0.397–0.532
	Male	k	0.378	0.350–0.411
	Female	$\hat{t}_0$	–0.012	–0.013 to –0.010
	Male	$\hat{t}_0$	–0.012	–0.013 to –0.011
	Female	σ	0.157	0.147–0.168
	Male	σ	0.108	0.101–0.115
	Female	τ	0.572	0.528–0.619
	Male	τ	0.528	0.490–0.569
Spine—new	Female	L_{∞}	447	431–464
	Male	L_{∞}	502	490–514
	Female	k	0.390	0.345–0.443
	Male	k	0.390	0.361–0.422
	Female	$\hat{t}_0$	–0.013	–0.014 to –0.012
	Male	$\hat{t}_0$	–0.012	–0.012 to –0.011
	Female	σ	0.147	0.137–0.158
	Male	σ	0.107	0.100–0.114
	Female	τ	0.536	0.497–0.581
	Male	τ	0.571	0.531–0.615
Spine—old	Female	L_{∞}	424	413–437
	Male	L_{∞}	486	476–498
	Female	k	0.598	0.531–0.675
	Male	k	0.507	0.467–0.548
	Female	$\hat{t}_0$	–0.009	–0.010 to –0.008
	Male	$\hat{t}_0$	–0.009	–0.010 to –0.009
	Female	σ	0.162	0.152–0.173
	Male	σ	0.129	0.121–0.137
	Female	τ	—	—
	Male	τ	—	—

Note: L_{∞} (mm) is the asymptotic length, k (year^{–1}) is the Brody growth coefficient, σ and τ are the likelihood standard deviations, and $\hat{t}_0$ is the derived age (years) at length zero. Only one reader was used for the old spine protocol, so there is no inter-reader variability (τ) estimate.

how changes in ageing protocols may affect derived parameter estimates or assessment outcomes. This is particularly salient in the context of ageing error and sexually dimorphic

growth. Ageing protocols are frequently amended when ageing error is detected; thus, the difference in growth estimates under an altered paradigm must be assessed. Additionally,

Fig. 4. Posterior distribution of the difference in asymptotic length (L_∞ ; A), the Brody growth coefficient (k ; B), the likelihood standard deviations (σ ; C), the likelihood standard deviation due to observation error (τ ; D), and size at length zero (t_0 ; E) estimates between males and females for otolith- (orange line), new spine protocol- (purple line), and old spine protocol- (blue line) derived age estimates. The maximum *a posteriori* (MAP) value, by ageing method, is reported for comparison between sex in the top left of each panel with significant differences assessed at $\alpha = 0.10$.



many fish exhibit sexually dimorphic growth necessitating the investigation of the effect of sex on growth parameter estimates beyond size-at-age. Bayesian methods can directly facilitate this comparison as the parameter estimates from different growth models can be simultaneously estimated. In our application, fitting sexually dimorphic growth models for each protocol allowed us to compare differences between sexes within an ageing protocol and between protocols. This ensures that significantly different comparisons take into account the joint distribution across all growth parameters. As L_∞ and k are highly correlated in the VBGM, the error in the joint distribution is more representative of the true error. One potential limitation of the approach described here is the use of weakly informative priors derived from the maximum likelihood parameter estimates (Gelman et al. 2017). However, a sensitivity run with hyper weakly informed priors, derived from Jefferson et al. (2019), revealed no effect on final parameter estimates demonstrating the approach is robust to central tendencies of the priors and defining the priors as we have had the added benefit of cutting computation time by two-thirds.

Bias in age estimates propagates as bias in growth parameter estimates, principally the overestimation of the Brody growth coefficient (k), which can result in the overestimation of M , and thus the overestimation of stock productivity. This is not exceptionally surprising as ageing error does not change the range of lengths included in a model. Therefore, ageing error primarily affects k and not the parameters more strongly influenced by the lengths included in the model like L_∞ and σ . The sex-specific VBGM estimates of k reported here, based on otolith-derived and new spine protocol age estimates, were significantly lower than k estimated here with the old spine protocol. Additionally, they are generally lower than estimates, based on the old dorsal spine protocol, reported in several previous studies of gray triggerfish in the nGOM and southeastern Atlantic ($k = 0.52\text{--}0.95 \text{ year}^{-1}$; Allman et al. 2018; Kelly-Stormer et al. 2017). These combined results support the conclusion that estimates of k will be elevated, and growth overestimated, if age is estimated using the Kolmos et al. (2013) protocol.

Several authors have reported estimates of k , derived from whole otolith opaque zone counts, for Atlantic gray trigger-

fish that are lower than those reported here ($k = 0.27\text{--}0.30$ year⁻¹; Shervette et al. 2021; Shervette and Rivera Hernández 2022). However, VBGMs in those studies were fit to data sets that have a distinct lack of young, small fish in the 50–250 mm range due to gear selectivity (Shervette et al. 2021; Shervette and Rivera Hernández 2022). This manifests in the negative t_0 estimates ranging from -0.30 to -0.19 years, which results in L_0 estimates of 22.6 to 36.2 mm, 10 to 16 times larger than the observed L_0 (Simmons and Szedlmayer 2013; Shervette et al. 2021; Shervette and Rivera Hernández 2022). Size-selective sampling can result in a dampening of the estimate of k ; thus, growth appears to occur at a slower rate than would be estimated if the samples were representative of the population (Gwinn et al. 2010). The incorporation of additional information, such as an informative prior on L_0 , in a Bayesian framework can reduce biases in VBGM estimates due to size-selective sampling.

Ageing error and growth parameter misspecification can result in misestimation of stock status benchmarks, increasing the likelihood of overfishing (Reeves 2003; Stawitz et al. 2019). Accurate estimation of growth parameters is thus vital to effective fisheries management. Here, we have demonstrated the utility of a novel modeling approach for estimating and statistically comparing growth parameters across sex and multiple ageing protocols while incorporating multiple readers. This approach explicitly acknowledges and quantifies uncertainty due to reader imprecision, is broadly applicable within growth estimation, and provides a framework that can be expanded to incorporate additional sources of uncertainty such as measurement error. Beyond this we have demonstrated gray triggerfish otoliths and the newly proposed and validated dorsal spine protocol (Potts et al. 2023; Chamberlin et al. 2024) produce similar growth estimates while the old dorsal spine protocol (Kolmos et al. 2013) overestimates k , erroneously indicating a more productive stock.

Acknowledgements

The authors would like to thank the many researchers and fishers who were essential to collecting and processing the data presented here. In particular, we would like to acknowledge Joe Tarnecki and Brett Falterman for aid in collecting samples. The scientific results and conclusions, as well as any views or opinions expressed herein, are those of the author(s) and do not necessarily reflect those of NOAA or the U.S. Department of Commerce.

Article information

History dates

Received: 10 October 2024

Accepted: 14 May 2025

Accepted manuscript online: 30 May 2025

Version of record online: 7 July 2025

Copyright

© 2025 Authors Siders, Rogers, Taylor, and Patterson. This work is licensed under a [Creative Commons Attribution 4.0](https://creativecommons.org/licenses/by/4.0/)

International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author(s) and source are credited.

Data availability

Data generated or analyzed during this study are available at <https://doi.org/10.17605/OSF.IO/5ZMFQ>.

Author information

Author ORCIDs

Derek W. Chamberlin <https://orcid.org/0000-0002-4270-7245>

Author contributions

Conceptualization: DWC, ZAS, JCP, WDR, WFP

Formal analysis: DWC, ZAS, WFP

Funding acquisition: WFP

Investigation: DWC, JCP, WDR, WFP

Methodology: DWC, ZAS, JCP, WDR, WFP

Project administration: MAT

Resources: MAT

Supervision: WFP

Visualization: DWC, ZAS

Writing – original draft: DWC, ZAS

Writing – review & editing: JCP, WDR, MAT, WFP

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding information

Funding for this study was provided by the Gulf of Mexico Fishery Management Council (NA20NMF4410007) and NOAA Marine Fisheries Initiative (NA21NMF4330501) to WFP. Additional funding was provided by the University of Florida.

Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/cjfas-2024-0315>.

References

- Allman, R.J., Fioramonti, C.L., Patterson, W.F., and Pacicco, A.E. 2016. Validation of annual growth-zone formation in gray triggerfish *Balistes capriscus* dorsal spines, fin rays, and vertebrae. *Gulf Mex. Sci.* **33**: 68–67. doi:[10.18785/goms.3301.06](https://doi.org/10.18785/goms.3301.06).
- Allman, R.J., Patterson, W.F., Fioramonti, C.L., and Pacicco, A.E. 2018. Factors affecting estimates of size at age and growth in grey triggerfish *Balistes capriscus* from the northern Gulf of Mexico. *J. Fish Biol.* **92**: 386–398. doi:[10.1111/jfb.13518](https://doi.org/10.1111/jfb.13518). PMID: 29243251.
- Beamish, R.J., and Fournier, D.A. 1981. A method for comparing the precision of a set of age determinations. *Can. J. Fish. Aquat. Sci.* **38**: 982–983. doi:[10.1139/f81-132](https://doi.org/10.1139/f81-132).
- Beamish, R.J., and McFarlane, G.A. 1983. The forgotten requirement for age validation in fisheries biology. *Trans. Am. Fish. Soc.* **112**: 735–743. doi:[10.1577/1548-8659\(1983\)112\(735:TFRAV\)2.0.CO;2](https://doi.org/10.1577/1548-8659(1983)112(735:TFRAV)2.0.CO;2).
- Bunch, A.J., Walters, C.J., and Coggin, L.G., Jr. 2013. Measurement error in fish lengths: evaluation and management implications. *Fisheries*, **38**(7): 320–326. doi:[10.1080/03632415.2013.803472](https://doi.org/10.1080/03632415.2013.803472).

- Bradford, M.J. 1991. Effects of ageing errors on recruitment time series estimated from sequential population analysis. *Can. J. Fish. Aquat. Sci.* **48**: 555–558. doi:10.1139/f91-070.
- Caltabellotta, F.P., Siders, Z.A., Murie, D.J., Motta, F.S., Cailliet, G.M., and Gadig, O.B.F. 2019. Age and growth of three endemic threatened guitarfishes *Pseudobatos horkelii*, *P. percellens* and *Zapteryx brevirostris* in the western South Atlantic Ocean. *J. Fish Biol.* **95**: 1236–1248. doi:10.1111/jfb.14123. PMID: 31429078.
- Caltabellotta, F.P., Siders, Z.A., Cailliet, G.M., Motta, F.S., and Gadig, O.B.F. 2021. Preliminary age and growth of the deep-water goblin shark *Mitsukurina owstoni* (Jordan, 1898). *Mar. Freshwater Res.* **72**: 432. doi:10.1071/MF19370.
- Campana, S.E. 2001. Accuracy, precision and quality control in age determination, including a review of the use and abuse of age validation methods. *J. Fish Biol.* **59**: 197–242. doi:10.1111/j.1095-8649.2001.tb00127.x.
- Campana, S.E., Annand, M.C., and Mcmillan, A.I. 1995. Graphical and statistical methods for determining the consistency of age determinations. *Trans. Am. Fish. Soc.* **124**: 131–138. doi:10.1577/1548-8659(1995)124<0131:GASMF2.3.CO;2.
- Chamberlin, D.W., Siders, Z.A., Barnett, B.K., Ahrens, R.N.M., and Patterson, W.F. III 2023a. Highly variable length-at-age in vermilion snapper (*Rhomboplites aurorubens*) validated via Bayesian analysis of bomb radiocarbon. *Fish. Res.* **264**: 1–9. doi:10.1016/j.fishres.2023.106732.
- Chamberlin, D.W., Siders, Z.A., Barnett, B.K., and Patterson, W.F. III 2023b. Eye lens-derived $\Delta^{14}\text{C}$ signatures validate extreme longevity in the deepwater scorpaenid blackbelly rosefish (*Helicolenus dactylopterus*). *Sci. Rep.* **13**: 1–11. doi:10.1038/s41598-023-34680-0.
- Chamberlin, D.W., Potts, J.C., Rogers, W.D., Siders, Z.A., and Patterson, W.F. III 2024. Bomb ^{14}C validates gray triggerfish (*Balistes caprisus*) dorsal spine and otolith ageing protocols. *Fish. Res.* **279**: 1–10. doi:10.1016/j.fishres.2024.107123.
- Cope, J.M., and Punt, A.E. 2007. Admitting ageing error when fitting growth curves: an example using the von Bertalanffy growth function with random effects. *Can. J. Fish. Aquat. Sci.* **64**: 205–218. doi:10.1139/f06-179.
- Fisch, N.C., Bence, J.R., Myers, J.T., Berglund, E.K., and Yule, D.L. 2019. A comparison of age- and size-structured assessment models applied to a stock of cisco in Thunder Bay, Ontario. *Fish. Res.* **209**: 86–100. doi:10.1016/j.fishres.2018.09.014.
- Gabry, J., and Češnovar, R. 2022. cmdstanr: R interface to “CmdStan”.
- Gelman, A., and Rubin, D.B. 1992. Inference from iterative simulation using multiple sequences. *Stat. Sci.* **7**. doi:10.1214/ss/1177011136.
- Gelman, A., Simpson, D., and Betancourt, M. 2017. The prior can often only be understood in the context of the likelihood. *Entropy*, **19**: 555. doi:10.3390/e19100555.
- Gwinn, D.C., Allen, M.S., and Rogers, M.W. 2010. Evaluation of procedures to reduce bias in fish growth parameter estimates resulting from size-selective sampling. *Fish. Res.* **105**: 75–79. doi:10.1016/j.fishres.2010.03.005.
- Ingram, W.G. 2001. Stock structure of gray triggerfish, *Balistes caprisus*, on multiple spatial scales in the Gulf of Mexico. University of South Alabama, Mobile, AL. 242pp.
- Jefferson, A.E., Allman, R.J., Pacicco, A.E., Franks, J.S., Hernandez, F.J., Albins, M.A., et al. 2019. Age and growth of gray triggerfish (*Balistes caprisus*) from a north-central Gulf of Mexico artificial reef zone. *Bull. Mar. Sci.* **95**: 177–195. doi:10.5343/bms.2018.0025.
- Kelly-Stormer, A., Shervette, V., Kolmos, K., Wyanski, D., Smart, T., McDonough, C., and Reichert, M.J.M. 2017. Gray triggerfish reproductive biology, age, and growth off the Atlantic coast of the southeastern USA. *Trans. Am. Fish. Soc.* **146**: 523–538. doi:10.1080/00028487.2017.1281165.
- Kolmos, K., Ballender, J., and Shervette, V. 2013. Report on age determination and reproductive classification workshops for gray triggerfish (*Balistes caprisus*). September 2011 and October 2012 42.
- Lin, Lai, H., and Gunderson, D.R. 1987. Effects of ageing errors on estimates of growth, mortality and yield per recruit for walleye pollock (*Theragra chalcogramma*). *Fish. Res.* **5**: 287–302. doi:10.1016/0165-7836(87)90048-8.
- Makowski, D., Ben-Shachar, M.S., Annabel Chen, S.H., and Ldecke, D. 2019a. Indices of effect existence and significance in the Bayesian framework. *Front. Psychol.* **10**: 2767. doi:10.3389/fpsyg.2019.02767.
- Makowski, D., Ben-Shachar, M., and Lüdtke, D. 2019b. bayestestR: describing effects and their uncertainty, existence and significance within the Bayesian framework. *J. Open Source Softw.* **4**: 1541. doi:10.21105/joss.01541.
- Methot, R.D., Wetzel, C.R., Taylor, I.G., and Doering, K. 2020. Stock synthesis user manual : version 3.30.15. doi:10.25923/5WPN-QT71.
- Moore, J.L. 2001. Age, growth, and reproductive biology of the gray triggerfish (*Balistes caprisus*) from the southeastern United States, 1992–1997. University of Charleston. 99pp.
- Pauly, D. 1980. On the interrelationships between natural mortality, growth parameters, and mean environmental temperature in 175 fish stocks. *ICES J. Mar. Sci.* **39**: 175–192. doi:10.1093/icesjms/39.2.175.
- Potts, J.C., Rogers, W.D., Rezek, T.C., and Rezek, A.R. 2023. Validation of annual growth zone formation in gray triggerfish *Balistes caprisus* dorsal spines, vertebrae, and otoliths. *Fish. Res.* **267**. doi:10.1016/j.fishres.2023.106809.
- Quinn, T.J., II, and Deriso, R.B. 1999. Quantitative fish dynamics. Oxford University Press. doi:10.1093/oso/9780195076318.001.0001.
- R Core Team. 2023. R: a language and environment for statistical computing.
- Reeves, S. 2003. A simulation study of the implications of age-reading errors for stock assessment and management advice. *ICES J. Mar. Sci.* **60**: 314–328. doi:10.1016/S1054-3139(03)00011-0.
- Rolim, F.A., Siders, Z.A., Caltabellotta, F.P., Rotundo, M.M., and Vaske-Júnior, T. 2020. Growth and derived life-history characteristics of the Brazilian electric ray *Narcine brasiliensis*. *J. Fish Biol.* **97**: 396–408. doi:10.1111/jfb.14378. PMID: 32402101.
- SEDAR. 2006. SEDAR 9 Stock Assessment Report, Gulf of Mexico Gray Triggerfish. Southeast, Data, Assessment, and Review, Charleston, South Carolina. Available from <https://sedarweb.org/documents/sedar-09-stock-assessment-report-gulf-of-mexico-gray-triggerfish/> [accessed December 2018].
- SEDAR. 2015. Stock Assessment Review Summary: Gulf Gray Triggerfish. Southeast, Data, Assessment, and Review, Charleston, South Carolina. Available from <https://sedarweb.org/documents/sedar-43-final-stock-assessment-report-gulf-of-mexico-gray-triggerfish/> [accessed December 2018].
- Shepard, K.E., Patterson, W.F. III, DeVries, D.A., and Ortiz, M. 2010. Contemporary versus historical estimates of king mackerel (*Scomberomorus cavalla*) age and growth in the U.S. Atlantic and Gulf of Mexico. *Bull. Mar. Sci.* **86**: 515–532.
- Shervette, V.R., Rivera Hernández, J.M., and Nunoo, F.K.E. 2021. Age and growth of grey triggerfish *Balistes caprisus* from trans-atlantic populations. *J. Fish Biol.* **98**: 1120–1136. doi:10.1111/jfb.14644. PMID: 33314115.
- Shervette, V.R., and Rivera Hernández, J.M. 2022. Illuminating otoliths: new insights for life history of *Balistes* triggerfishes. *J. Fish Biol.* **101**: 1557–1568. doi:10.1111/jfb.15233. PMID: 36181378.
- Siders, Z.A., Stratmann, T.A., Turner Tomaszewicz, C.N., Walde, A.S., and Munscher, E.C. 2023. Somatic growth and maturity for four species of river cooter including *Pseudemys concinna suwanniensis*, *P. nelsoni*, *P. peninsularis*, and *P. texana*. *Biology*, **12**: 965. doi:10.3390/biology12070965. PMID: 37508395.
- Simmons, C.M., and Szedlmayer, S.T. 2013. Description of reared pre-flexion gray triggerfish, *Balistes Caprisus*, larvae from the northern Gulf of Mexico. *Bull. Mar. Sci.* **89**: 643–652. doi:10.5343/bms.2012.1062.
- Stan Development Team. 2022. RStan: the R interface to Stan.
- Stawitz, C.C., Haltuch, M.A., and Johnson, K.F. 2019. How does growth misspecification affect management advice derived from an integrated fisheries stock assessment model? *Fish. Res.* **213**: 12–21. doi:10.1016/j.fishres.2019.01.004.
- Tyler, A.V., Beamish, R.J., and McFarlane, G.A. 1989. Implications of age determination errors to yield estimate. In Effects of ocean variability on recruitment and an evaluation of parameters used in stock assessment models. Edited by R.J. Beamish and G.A. McFarlane. Canadian Special Publication of Fisheries and Aquatic Sciences. pp. 27–35.
- von Bertalanffy, L. 1934. Untersuchungen über die gesetzlichkeit des Wachstums: I. Teil: allgemeine Grundlagen der Theorie; mathematische und physiologische gesetzlichkeiten des Wachstums bei

- Wassertieren. Wilhelm Roux' Arch. Entwickl. Org. **131**: 613–652. doi:[10.1007/BF00650112](https://doi.org/10.1007/BF00650112).
- Walters, C.J., and Martell, S.J.D. 2004. Fisheries ecology and management. Princeton University Press, Princeton, NJ.
- Wilson, K.L., Matthias, B.G., Barbour, A.B., Ahrens, R.N.M., Tuten, T., and Allen, M.S. 2015. Combining samples from multiple gears helps to avoid fishy growth curves. North Am. J. Fish. Manage. **35**: 1121–1131. doi:[10.1080/02755947.2015.1079573](https://doi.org/10.1080/02755947.2015.1079573).
- Zhou, S., Yin, S., Thorson, J.T., Smith, A.D.M., and Fuller, M. 2012. Linking fishing mortality reference points to life history traits: an empirical study. Can. J. Fish. Aquat. Sci. **69**: 1292–1301. doi:[10.1139/f2012-060](https://doi.org/10.1139/f2012-060).